

Hao Vo

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EDUCATION

University of Arkansas

Master, Computer Science - advised by [Dr. Ngan Le](#)

AR, US

May 2025 – May 2027

University of Information Technology - VNUHCM

B.S., Computer Science

HCM, VN

Aug. 2021 – Dec. 2024

RESEARCH INTERESTS

My research focuses on **4D scene understanding** - injecting geometric and temporal awareness into models from multi-view images and videos. My goal is to lift model intelligence from the image plane into space and time, enabling applications in robotics and autonomous driving.

SELECTED PUBLICATIONS

Hao Vo et al. "SemLT3D: Semantic-Guided Expert Distillation for Camera-only Long-Tailed 3D Object Detection." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**). 2026.

Hao Vo et al. "DRIVESPATIAL: A Benchmark for Spatiotemporal Intelligence in VLMs for Autonomous Driving." Under Review. 2026.

Hao Vo et al. "Robust motorcycle helmet detection in real-world scenarios: Using co-detr and minority class enhancement." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPRW**). 2024.

Thanh Phan, **Hao Vo** et al. "MVDGC: Joint 3D and 2D Multi-view Pedestrian Detection via Dual Geometric Constraints" In Transactions on Machine Learning Research (**TMLR**). 2026.

EXPERIENCE

Graduate Research Assistant

May. 2025 - May. 2027

University of Arkansas

AR, US

- Research **Multi-view Understanding** and **Spatial Intelligence** for autonomous systems, spanning *Autonomous Driving* and *Robotics*. Authored papers published and under submission at top-tier venues including **CVPR**, **NeurIPS**, and **TMLR**.

AI Engineer

Sep. 2023 - Apr. 2025

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Hong Kong

- Developed computer vision algorithms for safety-focused construction analytics. Led an **8-member** AI deployment team across Singapore & KSA, managing **20+ projects** and **300+ cameras**; built MLOps tooling (Airflow, Ansible, auto-labeling) cutting data-processing time by **30%**; optimized inference to serve **10 cameras** at real-time FPS on Jetson NX.

Undergraduate Research Assistant

Sep. 2021 - May. 2023

University of Information Technology - VNUHCM

HCM, VN

- Large-scale Video Retrieval**: Designed and developed a high-performance system capable of retrieving a ~10-second event from **1.3 TB** of video data in just **0.61s**, using flexible input modalities (e.g., speech, text, image). This system won the largest AI competition in Vietnam.

AWARDS

Second Prize in Vietnam Ministerial-level Scientific Research. 2025.

First Prize of AI City Challenge (Track 5), **NVIDIA**. 2024.

Second Prize FPT Vietnam AI Challenge. 2023.

First Prize Vietnam HCM AI Challenge. 2022.

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++, Go.

Frameworks: Torch, React, Node.js, Next.js, FastAPI.

Developer Tools: Git, Docker, Airflow, Ansible, Vault, Signoz, MLflow, Celery, Redis, RabbitMQ, Nginx, Slurm, AWS, GitCI, MongoDB, SQL, DVC.